고완풀 단원별 적분 1단원 해설

[1] $\displaystyle\int_0^{\tanh^{-1}\frac{1}{\sqrt5}} \frac{\sinh x\cosh x}{\sinh^4 x+\cosh^4 x}dx$

$\displaystyle=\int_0^{\tanh^{-1}\frac{1}{\sqrt5}} \frac{\frac{\sinh x}{\cosh x}\times\frac{1}{\cosh^2 x}}{\frac{\sinh^4 x}{\cosh^4 x}+1}dx$

$\displaystyle=\int_0^{\tanh^{-1}\frac{1}{\sqrt5}} \frac{\tanh x\times \operatorname{sech}^2 x}{\tanh^4 x+1}dx$

$\displaystyle=\int_0^{\frac{1}{\sqrt5}} \frac{t}{t^4+1}dt\ \ (\because \tanh x=t)$

$\displaystyle=\frac12\int_0^{\frac{1}{\sqrt5}} \frac{2t}{(t^2)^2+1}dt=\frac12\left[\tan^{-1}(t^2)\right]_0^{\frac{1}{\sqrt5}}$

$\displaystyle=\frac12\tan^{-1}\frac15=\alpha$

이므로 $\tan 2\alpha=\frac15$이고, $\cos 2\alpha=\frac{5}{\sqrt{26}}$이다.

즉, $\cos 4\alpha=2\cos^2 2\alpha-1=\frac{12}{13}$

정답 : ③

[2] $\displaystyle\tan^{-1}m_1-\tan^{-1}m_2=\tan^{-1}\frac{m_1-m_2}{1+m_1m_2}$이므로

$\displaystyle\int_0^3 \tan^{-1}\frac{3-x}{1+3x}dx=\int_0^3 \left(\tan^{-1}3-\tan^{-1}x\right)dx$

$\displaystyle=3\tan^{-1}3-\left[x\tan^{-1}x-\int\frac{x}{1+x^2}dx\right]_0^3$

$\displaystyle=3\tan^{-1}3-\left[x\tan^{-1}x-\frac12\ln(1+x^2)\right]_0^3$

$\displaystyle=3\tan^{-1}3-3\tan^{-1}3+\frac12\ln 10$

$\displaystyle=\frac12\ln 10$ 이다.

정답 : ①

[3] $\displaystyle g(a)+g\left(\frac1a\right)=\int_1^a\frac{\ln t}{1+t}dt+\int_1^{\frac1a}\frac{\ln t}{1+t}dt$

$\displaystyle=\int_1^a\frac{\ln t}{1+t}dt+\int_1^a\frac{\ln\frac1x}{1+\frac1x}\times-\frac{1}{x^2}dx\ \left(\because t=\frac1x\right)$

$\displaystyle=\int_1^a\frac{\ln t}{1+t}dt+\int_1^a\frac{\ln x}{x(x+1)}dx$

$\displaystyle=\int_1^a\ln t\times\frac{1+t}{t(t+1)}dt=\int_1^a\frac{\ln t}{t}dt$

$\displaystyle=\frac12\left[(\ln t)^2\right]_1^a=\frac12(\ln a)^2$

정답 : ⑤

[4] $\displaystyle\int_1^2\frac{1}{x^2}\sqrt{\frac{x-1}{x+1}}dx=\int_1^2\frac{1}{x^2}\sqrt{\frac{(x-1)^2}{(x+1)(x-1)}}dx$

$\displaystyle=\int_1^2\frac{x-1}{x^2\sqrt{x^2-1}}dx$

$\displaystyle=\int_0^{\frac{\pi}{3}}\frac{(\sec\theta-1)\sec\theta\tan\theta}{\sec^2\theta\tan\theta}d\theta\ \ (\because x=\sec\theta)$

$\displaystyle=\int_0^{\frac{\pi}{3}}\frac{\sec\theta-1}{\sec\theta}d\theta=\int_0^{\frac{\pi}{3}}(1-\cos\theta)d\theta$

$\displaystyle=\left[\theta-\sin\theta\right]_0^{\pi/3}=\frac{\pi}{3}-\frac{\sqrt3}{2}$

정답 : ①

[5] $\displaystyle\int_0^1\frac{1+\tanh x}{\sqrt[3]{1-\tanh x}}dx=\int_0^1\frac{(1+\tanh x)(1-\tanh x)}{(1-\tanh x)^{\frac43}}dx$

$\displaystyle=\int_0^1\frac{1-\tanh^2 x}{(1-\tanh x)^{\frac43}}dx=\int_0^1\operatorname{sech}^2 x\,(1-\tanh x)^{-\frac43}dx$

$\displaystyle=3\left[(1-\tanh x)^{-\frac13}\right]_0^1=3\left(\frac{1}{\sqrt[3]{1-\tanh 1}}-1\right)$

정답 : ⑤

[6] $\displaystyle\frac{1-x+2x^2-x^3}{x(x^2+1)^2}$

$\displaystyle=\frac1x+\frac{ax+b}{x^2+1}+\frac{cx+d}{(x^2+1)^2}$

$\displaystyle=\frac{(x^2+1)^2+x(ax+b)(x^2+1)+x(cx+d)}{(x^2+1)^2}$

$\displaystyle=\frac{(1+a)x^4+bx^3+(2+a+c)x^2+(b+d)x+1}{(x^2+1)^2}$

이므로 분자를 비교하면 $a=-1, b=-1, c=1, d=0$이다.

$\displaystyle\int_1^{\sqrt3}\frac{1-x+2x^2-x^3}{x(x^2+1)^2}dx=\int_1^{\sqrt3}\left(\frac1x+\frac{-x-1}{x^2+1}+\frac{x}{(x^2+1)^2}\right)dx$

$\displaystyle=\left[\ln x-\frac12\ln(x^2+1)-\tan^{-1}x-\frac{1}{2(x^2+1)}\right]_1^{\sqrt3}$

$\displaystyle=\ln\sqrt3-\frac12\ln 4-\frac{\pi}{3}-\frac18+\frac12\ln 2+\frac{\pi}{4}+\frac14$

$\displaystyle=\frac12\ln 3-\frac12\ln 2-\frac{\pi}{12}+\frac18$

$\displaystyle=\frac12\ln\frac32-\frac{\pi}{12}+\frac18$

정답 : ④

[7] $\displaystyle F(x)=\frac{2\sqrt3}{3}\tan^{-1}\left(\frac{\alpha x}{1+\beta x^2}\right)+C$일 때,

$\displaystyle F'(x)=\frac{2\sqrt3}{3}\times\frac{1}{1+\left(\frac{\alpha x}{1+\beta x^2}\right)^2}\times\frac{\alpha(1+\beta x^2)-2\alpha\beta x^2}{(1+\beta x^2)^2}$

$\displaystyle=\frac{2\sqrt3}{3}\times\frac{\alpha(1+\beta x^2)-2\alpha\beta x^2}{(1+\beta x^2)^2+(\alpha x)^2}=\frac{2(x^2+1)}{x^4+x^2+1}$

$\displaystyle\Rightarrow \frac{2\sqrt3}{3}\alpha=2$이므로 $\alpha=\sqrt3$이다.

$\displaystyle\Rightarrow \frac{2\sqrt3}{3}\times\frac{\sqrt3(1-\beta x^2)}{\beta^2x^4+2\beta x^2+1+3x^2}=\frac{2(x^2+1)}{x^4+x^2+1}$이므로

$\beta=-1$이다.

즉, $\alpha+\beta=\sqrt3-1$

[다른 풀이]

$\displaystyle\int\frac{2(x^2+1)}{x^4+x^2+1}dx=\int\frac{2(x^2+1)}{x^4-2x^2+1+3x^2}dx$

$$= \int \frac{2(x^2+1)}{(x^2-1)^2+(\sqrt{3}x)^2}dx = \int \frac{\frac{2(x^2+1)}{(x^2-1)^2}}{1+\left(\frac{\sqrt{3}x}{x^2-1}\right)^2}dx$$

$$= \int \frac{2}{1+t^2}\left(-\frac{1}{\sqrt{3}}\right)dt \quad \left(\because \frac{\sqrt{3}x}{x^2-1}=t\right)$$

$$= -\frac{2}{\sqrt{3}}\tan^{-1}t + C = -\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{\sqrt{3}x}{x^2-1}\right)+C$$

$= \frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{\sqrt{3}x}{1-x^2}\right)+C$이므로 $\alpha=\sqrt{3}$, $\beta=-1$ 이다.

$\therefore \alpha+\beta=\sqrt{3}-1$

정답 : ③

【8】 (가) $\int \frac{2x}{x^2-1}dx = \int \frac{2x}{(x+1)(x-1)}dx = \int \frac{1}{x+1}+\frac{1}{x-1}dx$

(나) $\int \frac{2}{x(x+1)}dx = \int \frac{2}{x}-\frac{2}{x+1}dx$

(다) $\int \frac{x^2+1}{x^2(x+1)^2}dx = \int \frac{-2}{x}+\frac{1}{x^2}+\frac{2}{x+1}+\frac{2}{(x+1)^2}dx$

(라) $\int \frac{x^2+1}{x^2(x+1)}dx = \int \frac{-1}{x}+\frac{1}{x^2}+\frac{2}{x+1}dx$

이므로 유리함수로 표현할 수 있는 것은 (나)와 (다) 조합뿐이다.

정답 : ②

【9】 $I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x dx$이면 $I_{n+2} = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^{n+2} x dx$이므로

$$I_n + I_{n+2} = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^{n+2} x dx$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x + \cot^{n+2} x dx$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x(1+\cot^2 x)dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x \csc^2 x dx$$

$$= -\int_1^0 t^n dt \quad (\because \cot x = t)$$

$= -\left[\frac{1}{n+1}t^{n+1}\right]_1^0 = \frac{1}{n+1}$ 이다.

즉, $I_{98}+I_{100} = \frac{1}{99}$

정답 : ③

【10】 $I_{n+2} = \int_0^\pi \frac{\sin(n+2)x}{\sin x}dx = \int_0^\pi \frac{\sin[(n+1)x+x]}{\sin x}dx$

$$= \int_0^\pi \frac{\sin(n+1)x\cos x + \cos(n+1)x\sin x}{\sin x}dx$$

$$= \int_0^\pi \frac{\sin(n+1)x\cos x}{\sin x}dx + \int_0^\pi \cos(n+1)x dx$$

$$= \int_0^\pi \frac{\sin(n+1)x\cos x}{\sin x}dx + [\sin(n+1)x]_0^\pi$$

$$= \frac{1}{2}\int_0^\pi \frac{\sin(n+2)x+\sin(nx)}{\sin x}dx$$

$$= \frac{1}{2}\int_0^\pi \frac{\sin(n+2)x}{\sin x}dx + \frac{1}{2}\int_0^\pi \frac{\sin(nx)}{\sin x}dx$$

$$= \frac{1}{2}I_{n+2} + \frac{1}{2}I_n$$

따라서 $I_{n+2}=I_n$ 을 만족한다.

또한 $I_1 = \int_0^\pi \frac{\sin x}{\sin x}dx = \int_0^\pi 1 dx = \pi$ 이므로

$I_1 = I_3 = I_5 = \cdots = \pi$이고

$I_2 = \int_0^\pi \frac{\sin 2x}{\sin x}dx = \int_0^\pi \frac{2\sin x\cos x}{\sin x}dx = [\sin x]_0^\pi = 0$이므로

$I_2 = I_4 = I_6 = \cdots = 0$이다.

즉, $\int_0^\pi \frac{\sin nx}{\sin x}dx = \begin{cases} \pi & , n\text{이 홀수} \\ 0 & , n\text{이 짝수} \end{cases}$ 이다.

정답 : ③